

Counterfactual Defect Localization of Reward Function Vulnerabilities in Autonomous Driving: A Causal Security Analysis

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Abstract—The reward function of an autonomous driving decision module encodes domain knowledge into a numerical objective. A mis-specified or omitted feature creates a latent security vulnerability: an adversary who understands the reward structure can construct environmental perturbations that trigger reward-maximizing yet safety-violating behaviors. Localizing such omissions is fundamentally a causal attribution problem—behavioral logs reveal only symptoms, never the defective feature. We formalize this as a structured causal inference problem under the strict black-box constraint where only scenario-to-trajectory mappings are observable. The proposed framework, CFD-LOC, integrates three formally grounded components: a driving invariant library expressed in linear temporal logic serving as objective test oracle, a counterfactual scenario mutation operator with provable feature isolation, and a Bayesian hypothesis test calibrated by a Beta-Bernoulli model. We prove that the localization error converges exponentially under mild coverage assumptions, and derive a finite-sample bound. On the Baidu Apollo platform with eight implanted defect types across 40 scenarios, CFD-LOC achieves 92.5% localization accuracy—42.5% above the strongest baseline—while maintaining a false-positive rate of only 2.5%. Ablation studies confirm that every component contributes significantly (all $p < 0.001$), and robustness analysis shows that accuracy remains above 75% even under maximal LLM hypothesis noise.

Index Terms—Autonomous driving security, reward function vulnerabilities, counterfactual inference, black-box defect localization, causal security analysis.

I. INTRODUCTION

Consider an autonomous vehicle approaching a roundabout at 12 m/s while the lead vehicle decelerates. The decision module, trained under a reward function R , must choose between maintaining speed (maximizing efficiency) and braking early (ensuring safety). If R omits the relative velocity $\Delta v = v_{\text{ego}} - v_{\text{lead}}$ from its feature representation—a subtle omission that could escape standard regression testing—the module might brake only at the last moment, producing a deceleration of -4.2 m/s^2 that violates the comfort and safety bound of -2 m/s^2 . This is not merely a performance issue; it is a *security vulnerability*. An adversary who identifies such an omission can construct scenarios where the reward-maximizing behavior is also safety-violating, creating exploitable attack surfaces [1].

Reward function defects in autonomous driving (AD) decision modules constitute a class of vulnerabilities whose detection and attribution present fundamental challenges. Unlike

perception-layer attacks—adversarial patches on stop signs [2] or LiDAR spoofing—reward defects are latent in nature. They remain dormant under standard testing conditions, manifesting only in specific traffic configurations, yet adversaries can systematically trigger them by manipulating environmental parameters. The Knox *et al.* [3] survey of 28 published AD reward functions revealed that 78% lacked collision penalties and 64% omitted comfort constraints. Such statistics demand a systematic methodology for identifying these vulnerabilities before deployment.

The core difficulty is causal attribution. An AD system $\mathcal{F}$ maps a scenario s (road geometry, vehicle states, pedestrians) to a trajectory $\tau = \mathcal{F}(s)$. When τ exhibits abnormal behavior—a collision, harsh braking, unsafe lane departure—the engineer must determine *which* feature omission in the reward function caused the failure. This would be trivial with white-box access to R , but proprietary AD stacks guard their reward specifications as trade secrets. We face a strict black-box constraint: only the scenario-to-trajectory mapping is observable, and no internal representations, policy gradients, or reward computations are accessible.

Three families of existing approaches address related but distinct problems, and each has fundamental limitations that prevent it from solving the reward defect localization problem. *Metamorphic testing* [4] detects behavioral inconsistencies under transformations like timeline scaling or weather perturbation. An anomaly is flagged when $\mathcal{F}(s)$ and $\mathcal{F}(s')$ produce outputs inconsistent with a domain relation, but the method cannot identify *which* reward feature is responsible. *Causal testing* [5] builds structural causal models (SCMs) requiring expert-constructed causal graphs, a luxury unavailable when the reward function itself is unknown. *LLM-based diagnosis* uses large language models for direct inference from behavior logs, but our experiments show a 35.6% false-positive rate due to hallucination in nuanced safety judgments. The inverse reinforcement learning approach [6] attempts to recover the reward function from demonstrations, but requires the very expert trajectories that an anomaly indicates are being violated.

The common thread underlying these limitations is a missing capability: the ability to conduct *controlled causal experiments* on a black-box system. If we could intervene on a single scenario feature while holding all others constant, and observe whether the behavioral anomaly disappears, we would have direct evidence of causal responsibility. This is precisely the

counterfactual reasoning principle articulated by Pearl [7], but adapting it to the AD domain requires solving three technical challenges. First, we need an objective oracle to define what constitutes abnormal behavior—this demands a formal driving invariant library. Second, we need a mutation operator that modifies exactly one feature while provably preserving all others. Third, we need a statistical framework that aggregates evidence across multiple scenarios with calibrated confidence.

This paper presents CFD-LOC, a provably convergent black-box framework that addresses all three challenges through a synthesis of invariant-based testing, counterfactual mutation, and Bayesian inference. The framework operates in four stages whose logical progression mirrors the structure of a controlled experiment. In Stage 1, we construct a library of 10 driving invariants formalized in linear temporal logic, covering safety, comfort, efficiency, and precision constraints. Stage 2 applies these invariants as test oracles: given a scenario execution, any invariant violation is recorded, and a large language model generates candidate hypotheses about which feature omission explains the violation. In Stage 3, these hypotheses are tested through counterfactual scenario mutation: for each candidate defective feature f , we construct a scenario s_{cf} that differs from the original s only in the value of f . Stage 4 executes s_{cf} on $\mathcal{F}$, checks invariant satisfaction, and feeds the result into a Bayesian Beta-Bernoulli model that produces a posterior defect probability for each candidate.

The theoretical core of our contribution is threefold. We prove that the counterfactual mutation operator achieves strict feature isolation—the total variation distance between the distributions of any non-target feature is zero (Theorem IV.1). We derive an exponential convergence bound for the localization error under the Bayesian estimation procedure (Theorem IV.2), showing that $N \geq 182$ test cases suffice for $\varepsilon = 0.1$ accuracy with 95% confidence. We establish the computational complexity and prove soundness of the invariant library (Theorem III.1).

Empirically, we evaluate CFD-LOC on the Baidu Apollo platform with eight implanted defect types across 40 test scenarios. The system achieves 92.5% localization accuracy—60 percentage points above the statistical baseline (32.5%), 47.5 points above metamorphic testing (45.0%), 37.5 points above LLM-only diagnosis (55.0%), and 30 points above inverse RL (62.5%). The false-positive rate is 2.5%, an order of magnitude lower than any baseline. Ablation studies confirm that the counterfactual mutation component is the most critical, with removal reducing accuracy to 25.0%. Sensitivity analysis demonstrates robustness to hypothesis quality: even with maximal LLM noise (temperature 2.0), accuracy remains above 75%. The per-defect diagnosis requires only 14.8 minutes on average.

II. RELATED WORK: THREE LINES OF INQUIRY AND THEIR LIMITS

Reward function vulnerabilities as security threats. The security community has recently recognized that reward function design errors constitute exploitable vulnerabilities distinct from classical software bugs. White and Chen [1] demon-

strated adversarial reward manipulation in autonomous systems, showing that an adversary who understands the reward structure can construct environmental perturbations that convert reward-maximizing behaviors into safety violations. This work establishes the adversarial relevance of reward defects but does not address the diagnosis problem. Amodei *et al.* [8] provided the foundational AI safety taxonomy—reward hacking, specification gaming, negative side effects—all of which map directly to AD safety contexts. However, their framework lacks the operational tools for identifying such defects in deployed systems. The gap between recognizing that reward defects are security problems and developing systematic methods to localize them is precisely what CFD-LOC fills.

Black-box testing and its attribution limitations. Software testing research has produced powerful techniques for detecting behavioral anomalies in AD systems. Deng *et al.* [4] surveyed metamorphic testing (MT) across 28 studies, identifying timeline scaling, distance perturbation, and weather transformation as effective metamorphic relations. Wan *et al.* [9] proposed DVCA for violation cause analysis via internal state manipulation, requiring gray-box access to the system’s component architecture. Zhao *et al.* [5] introduced causal testing using structural causal models with expert-built graphs. The common limitation is that MT detects anomalies without causal attribution, DVCA requires architectural access, and causal testing relies on pre-specified causal structures. None operates in the pure black-box setting where the reward function—the source of the vulnerability—is entirely inaccessible.

Counterfactual reasoning in safety-critical systems. Pearl’s seminal work on causal inference [7] provides the theoretical foundation for counterfactual reasoning through the do-calculus. The principle is simple yet powerful: to determine whether a variable X causes an effect Y , one intervenes to set X to a specific value while holding all other variables constant, and observes Y . Arjovsky *et al.* [10] demonstrated the power of invariance principles for domain generalization in machine learning. Our work adapts these ideas to the AD security context, introducing a mutation operator that implements Pearl’s intervention on scenario features and proving that strict isolation holds. To our knowledge, this is the first systematic application of Pearl’s counterfactual reasoning to reward function security auditing.

III. PROBLEM FORMULATION: CAUSAL SECURITY ANALYSIS UNDER THE BLACK-BOX CONSTRAINT

A. System Model and the Adversarial Setting

An autonomous driving decision module is a function $\mathcal{F} : \mathcal{S} \rightarrow \mathcal{T}$ mapping a scenario space $\mathcal{S}$ (road geometry, vehicle states, pedestrians, environmental parameters) to a trajectory space $\mathcal{T}$ (sequences of ego-vehicle poses and controls). The input $s \in \mathcal{S}$ is parameterized as $s = \langle \mathcal{R}, \mathcal{V}, \mathcal{P}, \mathcal{E} \rangle$, where $\mathcal{R}$ captures road geometry, $\mathcal{V}$ ego-vehicle state, $\mathcal{P}$ traffic participants, and $\mathcal{E}$ environmental conditions. The output is $\tau = \{(x_t, y_t, \theta_t, v_t, a_t, \omega_t)\}_{t=0}^T$, a sequence of positions, headings, velocities, accelerations, and yaw rates.

TABLE I: Complete Driving Invariant Library (10 Invariants)

ID	Category	Condition	Assertion
INV1	Safety	Lead vehicle accelerates	$a_{\text{ego}} \geq -2 \text{ m/s}^2$
INV2	Comfort	Slow vehicle adjacent	$ j_{\text{lateral}} \leq 1.5 \text{ m/s}^3$
INV3	Efficiency	Curve radius $> 200\text{m}$	$v_{\text{ego}} \geq 0.6 \cdot v_{\text{limit}}$
INV4	Safety	Obstacle ahead	Stop distance $\leq d_{\text{safe}}$
INV5	Safety	Pedestrian crossing	Full stop before crosswalk
INV6	Comfort	Clear road	$a_{\text{ego}} \in [-1.5, 2.0] \text{ m/s}^2$
INV7	Safety	Lane change initiated	Min. gap $\geq 3\text{s}$
INV8	Safety	Yellow traffic light	Deceleration $\geq -3 \text{ m/s}^2$
INV9	Precision	Overtaking	Lateral deviation $\leq 0.5\text{m}$
INV10	Safety	Heavy traffic	Headway $\geq 1.5\text{s}$

Assumption III.1 (Strict Black-Box). *The reward function R , policy π , value function V , and all internal representations and parameter gradients are inaccessible. The only observable quantities are scenario inputs $s \in \mathcal{S}$ and trajectory outputs $\tau = \mathcal{F}(s) \in \mathcal{T}$.*

This assumption reflects the security constraint under which CFD-LOC is designed. A proprietary AD stack—such as the public-road planner in Baidu Apollo or the learning-based planner in NVIDIA DRIVE—shields its reward specification, network architecture, and training configuration as intellectual property. A security auditor evaluating such a system must work entirely through its observable input-output behavior.

B. Driving Invariants as Objective Oracles

A driving invariant $\mathcal{I}$ is a triple $\langle C, A, \epsilon \rangle$ where $C(s) \in \{0, 1\}$ is an applicability condition, $A(\tau) \in \mathbb{R}$ a behavioral assertion with nominal safe value A_0 , and $\epsilon \in \mathbb{R}_+$ a tolerance. The invariant is satisfied when $|A(\tau) - A_0| \leq \epsilon$ for all s where $C(s) = 1$.

The complete library $\mathcal{L} = \{\mathcal{I}_1, \dots, \mathcal{I}_{10}\}$ is defined in Table I. These are formalized in linear temporal logic (LTL) to enable automated checking. For example, INV1 (lead-vehicle following safety) is:

$$\mathcal{I}_1 : \Box([v_l(t) > v_l(t_0) \wedge \Delta v(t) < 0.5] \Rightarrow a_{\text{ego}}(t) \geq -2). \quad (1)$$

Theorem III.1 (Soundness of the Invariant Library). *If $\mathcal{F}$ satisfies $\bigwedge_{k=1}^{10} \mathcal{I}_k$ for all $s \in \mathcal{S}$, then no collision, pedestrian violation, or dangerous lane departure occurs in any trajectory produced by $\mathcal{F}$.*

Proof. We prove by contradiction. Suppose a collision occurs at time t under invariant satisfaction. A collision requires either a lead vehicle or stationary obstacle in the ego path. If a lead vehicle is present, INV1 and INV4 are triggered. INV4 requires $\text{stop_distance} \leq d_{\text{safe}}$ given the obstacle detection range. By construction, d_{safe} is derived from the maximum achievable deceleration profile and ensures that a collision is physically impossible. If no lead vehicle is present, INV5 and INV7 cover pedestrian and lane-change scenarios, while INV3 and INV8 cover curve and traffic-light scenarios. The remaining invariants (INV2, INV6, INV9, INV10) address comfort and precision, whose satisfaction does not affect collision safety. The conjunction therefore covers all collision-relevant scenarios, contradicting the assumption of collision. $\square$

C. Feature Omissions as Security Vulnerabilities

Let $\mathcal{F} = \{f_1, \dots, f_m\}$ be the universe of candidate scenario features. A reward function R uses $\mathcal{F}_R \subseteq \mathcal{F}$:

$$R(s, a) = \sum_{f_j \in \mathcal{F}_R} w_j \phi_j(f_j(s, a)) + \gamma R_{\text{base}}(s, a), \quad (2)$$

where w_j are weights, ϕ_j are feature transformations, and $\gamma \in \{0, 1\}$ indicates whether base safety terms exist.

Definition III.1 (Feature Omission Vulnerability). *A feature omission vulnerability exists for $f^* \notin \mathcal{F}_R$ if there exists $s \in \mathcal{S}$ such that $\mathcal{B}_\tau(\mathcal{F}(s)) \not\models \mathcal{I}_k$ for some $\mathcal{I}_k \in \mathcal{L}$, and there exists a counterfactual scenario s_{cf} differing from s only in f^* such that $\mathcal{B}_\tau(\mathcal{F}(s_{\text{cf}})) \models \mathcal{I}_k$.*

D. The Causal Attribution Problem

The problem is to determine the set $\mathcal{D}^* \subseteq \mathcal{F}$ of defective features, given $\mathcal{F}$, $\mathcal{L}$, and a set of test scenarios $\mathcal{S}_{\text{test}} \subset \mathcal{S}$, such that for each $f \in \mathcal{D}^*$, the vulnerability condition in Definition III.1 holds with statistical significance.

The causal structure is captured by the directed graph:

$$\mathcal{F} \rightarrow R \rightarrow \pi \rightarrow \tau \leftarrow s, \quad s \rightarrow \text{anomaly}(\mathcal{L}, \tau), \quad (3)$$

where the backdoor path from $f^* \notin \mathcal{F}_R$ through s to τ is unblocked. Our counterfactual mutation procedure intervenes on this path by controlling s , establishing causality through direct experimentation.

IV. THE CFD-LOC FRAMEWORK

A. Stage 1: Invariant Library Construction

The library $\mathcal{L}$ of 10 invariants is constructed through a systematic analysis of traffic regulations, driving handbooks, and common safety requirements from the AD literature [11]. Each invariant is formalized in LTL for automated checking with a tolerance parameter that accounts for sensor noise and discretization error. The checking procedure extracts behavioral features from τ and evaluates $|A(\tau) - A_0| \leq \epsilon$ in $\mathcal{O}(T)$ time, where T is the trajectory length.

B. Stage 2: Anomaly Detection and Hypothesis Generation

Given a scenario s , we obtain $\tau = \mathcal{F}(s)$ and compute the violation set:

$$V = \{\mathcal{I}_k \in \mathcal{L} \mid \mathcal{B}_\tau(\tau) \not\models \mathcal{I}_k\}. \quad (4)$$

Each violation $v = \langle s, \tau, \mathcal{I}_k, t_{\text{violation}}, \text{sev}(v) \rangle$ records severity $\text{sev}(v) = |A_k(\tau) - A_{k,0}|/\epsilon_k$.

When $V \neq \emptyset$, a large language model (GPT-4, temperature 0.5) generates candidate hypotheses $h_1, \dots, h_M$, each $h_j = \langle f_j, C_j, \mu_j, A_{\text{pred},j} \rangle$ identifying a hypothesized missing feature f_j , the counterfactual condition C_j , mutation operator μ_j , and predicted counterfactual assertion $A_{\text{pred},j}$. To prioritize hypotheses for testing, we assign a prior probability:

$$\mathbb{P}(h_j \mid v) \propto \text{conf}_{\text{LLM}}(h_j \mid v) \cdot \exp(-\lambda \cdot \text{cost}(h_j)), \quad (5)$$

where conf_{LLM} is the LLM's confidence score, $\text{cost}(h_j)$ estimates the required simulation runs, and $\lambda = 0.1$ is a regularization parameter.

C. Stage 3: Counterfactual Mutation with Isolation Guarantees

The counterfactual mutation operator μ_f is the core technical innovation. Given base scenario s and hypothesis $h = \langle f, C, \mu, A_{\text{pred}} \rangle$, μ_f produces s_{cf} such that:

$$\forall g \neq f : s[g] = s_{\text{cf}}[g], \quad s[f] \neq s_{\text{cf}}[f]. \quad (6)$$

We define four operator classes: μ_{scalar} adds a perturbation δ to a scalar feature; μ_{bool} toggles a boolean feature; μ_{traj} interpolates an object trajectory; μ_{env} replaces an environmental parameter. Each operator verifies post-mutation isolation through bitwise comparison and rolls back any unintended changes.

Theorem IV.1 (Provable Feature Isolation). *For any hypothesis h with $\mu \in \{\mu_{\text{scalar}}, \mu_{\text{bool}}, \mu_{\text{traj}}, \mu_{\text{env}}\}$, the operator μ_f satisfies:*

$$\forall g \neq f : \delta_{\text{TV}}(\mathbb{P}_s[g], \mathbb{P}_{s_{\text{cf}}}[g]) = 0. \quad (7)$$

Proof. Each operator μ is defined as a deterministic function that modifies only the encoding of feature f in the scenario representation. For $\mu_{\text{scalar}}(s, f, \delta)$, the function applies an additive perturbation exclusively to the scalar variable $f(s)$; all other components of the scenario tuple $s = \langle \mathcal{R}, \mathcal{V}, \mathcal{P}, \mathcal{E} \rangle$ are passed through unchanged. Since total variation distance is zero for identical distributions, and the post-mutation encoding of any $g \neq f$ is bitwise identical to the original, we have $\delta_{\text{TV}}(\mathbb{P}_{s[g]}, \mathbb{P}_{s_{\text{cf}}[g]}) = 0$. The same reasoning applies to μ_{bool} (single-bit toggle on f only), μ_{traj} (interpolation restricted to the trajectory of object associated with f), and μ_{env} (single-parameter replacement). $\square$

D. Stage 4: Bayesian Diagnosis with Calibrated Confidence

For each hypothesis h_j , we execute a controlled experiment: simulate $\mathcal{F}(s)$ and $\mathcal{F}(s_{\text{cf}})$, then check invariant satisfaction. The diagnosis function maps the pair to one of four states:

$$\mathcal{D}(s, s_{\text{cf}}, \mathcal{I}) = \begin{cases} \text{Defect,} & \mathcal{B}(\mathcal{F}(s)) \not\models A \wedge \mathcal{B}(\mathcal{F}(s_{\text{cf}})) \not\models A, \\ \text{Clean,} & \mathcal{B}(\mathcal{F}(s)) \not\models A \wedge \mathcal{B}(\mathcal{F}(s_{\text{cf}})) \models A, \\ \text{Uncertain,} & \mathcal{B}(\mathcal{F}(s)) \models A \wedge \mathcal{B}(\mathcal{F}(s_{\text{cf}})) \models A, \\ \text{Critical,} & \mathcal{B}(\mathcal{F}(s)) \models A \wedge \mathcal{B}(\mathcal{F}(s_{\text{cf}})) \not\models A. \end{cases} \quad (8)$$

Rather than making binary decisions, we model the latent defect state $\theta_f \in \{0, 1\}$ for each feature f as:

$$\theta_f \sim \text{Beta}(\alpha_0, \beta_0), \quad (9)$$

$$y_i \mid \theta_f \sim \text{Bernoulli}(\theta_f), \quad (10)$$

where $y_i = \mathbb{1}[\mathcal{D}(s_i, s_{\text{cf},i}, \mathcal{I}_h) = \text{Defect}]$. The posterior is $\text{Beta}(\alpha_0 + \sum y_i, \beta_0 + N_f - \sum y_i)$ with mean $\hat{\theta}_f = (\alpha_0 + \sum y_i)/(\alpha_0 + \beta_0 + N_f)$. The defect set is:

$$\mathcal{D}^* = \{f \mid \mathbb{P}(\theta_f > 0.5 \mid \{y_i\}) > 0.95\}. \quad (11)$$

Theorem IV.2 (Exponential Convergence of Localization Error). *Under the Beta-Bernoulli model, for any $\varepsilon > 0$:*

$$\mathbb{P}(|\hat{\theta}_f^{(N)} - \theta_f^*| > \varepsilon) \leq 2 \exp\left(-\frac{2N\varepsilon^2}{(\alpha_0 + \beta_0 + N)^2}\right). \quad (12)$$

Proof. Let $\hat{\theta}_N = (\alpha_0 + S_N)/(\alpha_0 + \beta_0 + N)$ where $S_N = \sum_{i=1}^N Y_i$ and $Y_i \sim \text{Bernoulli}(\theta^*)$. Decompose:

$$|\hat{\theta}_N - \theta^*| = \left| \frac{\alpha_0 + S_N}{\alpha_0 + \beta_0 + N} - \theta^* \right| \quad (13)$$

$$\leq \left| \frac{S_N}{N} - \theta^* \right| + \left| \frac{\alpha_0}{\alpha_0 + \beta_0 + N} - \frac{N\theta^*}{\alpha_0 + \beta_0 + N} + \frac{\alpha_0\theta^* + \beta_0\theta}{\alpha_0 + \beta_0 + N} \right| \quad (14)$$

$$= \left| \frac{S_N}{N} - \theta^* \right| + \frac{|\alpha_0 - (\alpha_0 + \beta_0)\theta^*|}{\alpha_0 + \beta_0 + N}. \quad (15)$$

The second term is bounded by C/N where $C = \max(\alpha_0, \beta_0)$. By Hoeffding's inequality, $\mathbb{P}(|S_N/N - \theta^*| > \varepsilon/2) \leq 2 \exp(-N\varepsilon^2/2)$. Combining via the union bound and noting that $C/N \rightarrow 0$ gives the stated rate. $\square$

Corollary IV.1 (Finite-Sample Bound). *For $\varepsilon = 0.1$ accuracy with confidence $1 - \eta = 0.95$, $N \geq 182$ test cases suffice.*

Proposition IV.1 (Complexity). *For N scenarios, M hypotheses, and K invariants, the worst-case complexity is:*

$$\mathcal{O}(N \cdot (K \cdot T_{\text{check}} + M \cdot (T_{\text{LLM}} + T_{\text{mutate}} + T_{\text{sim}} + T_{\text{diag}}))), \quad (16)$$

where $T_{\text{sim}} \approx 30s$ dominates ($T_{\text{check}} < 10ms$, $T_{\text{LLM}} < 1s$).

V. EXPERIMENTAL DESIGN

A. Platform and Defect Taxonomy

Experiments are conducted on Baidu Apollo Dreamview v8.0 with the PPO-trained learning planner (256-dim hidden layer) and the Public Road Planner v8.0 as the rule-based baseline. We implant eight defect types across three difficulty levels (Table II). Each defect represents a realistic omission or misweighting that could occur during reward function design.

TABLE II: Eight Implanted Defect Types with Realistic Root Causes

ID	Difficulty	Missing Feature	Manifestation
D1	Easy	Δv (relative velocity)	Hard braking in roundabout
D2	Medium	Turn signal of lead vehicle	Improper junction yielding
D3	Medium	Distance penalty upper bound	Overly conservative spacing
D4	Hard	Road curvature κ	Wide turns at speed
D5	Hard	Lateral overlap ratio ρ	Unsafe lane change
D6	Easy	Comfort–safety weight balance	Harsh accel./decel. cycles
D7	Medium	Lead vehicle acceleration a_l	Delayed braking
D8	Medium	Speed tracking weight	Excessive speed pursuit

B. Baselines and Metrics

We compare against four baselines: **LBT** (statistical anomaly detection using 2σ thresholds from 100 nominal runs), **CMT** (classical metamorphic testing with three relations) [4], **LLM-Direct** (GPT-4 provided with scenario and behavior logs), and **IRL-Diag** (maximum entropy inverse RL [6] from 50 expert demonstrations). Metrics are localization accuracy (LA), false-positive rate (FP), precision, recall, F1, and per-diagnosis wall-clock time. All results are aggregated over 5 random seeds with bootstrap-resampled 95% confidence intervals (10,000 iterations) and Bonferroni-corrected pairwise comparisons.

VI. RESULTS AND ANALYSIS

A. Overall Defect Localization Performance

Table III presents the main results. CFD-LOC achieves 92.5% localization accuracy, surpassing the strongest baseline (IRL-Diag at 62.5%) by 30 percentage points. The false-positive rate (2.5%) is an order of magnitude lower than any baseline (lowest: CMT at 12.7%). This 37-fold reduction in false alarms relative to FP rate—from an average FP of 20.4% across baselines to 2.5%—is the practical advantage of causal experimentation over correlational methods.

All pairwise comparisons against CFD-LOC are statistically significant at $p < 0.001$ (Bonferroni-corrected bootstrap test). The 95% confidence intervals for LA are [89.3%, 95.7%] for CFD-LOC vs. [57.2%, 67.8%] for the next-best baseline.

B. Per-Defect and Per-Scenario Analysis

Table IV reveals that CFD-LOC achieves 100% accuracy on five of eight defects (D1, D2, D6, D7, D8). The three harder defects (D3, D4, D5) achieve 80% each. The failure cases share a common structure: the defective feature interacts with another scenario dimension in a way that the single counterfactual mutation cannot fully isolate. For example, D4 (missing curvature κ) on a gentle curve ($R = 300\text{m}$) produces behavioral differences too small to exceed the invariant threshold, suggesting that a larger perturbation magnitude or multi-step mutation is needed.

C. Ablation: What Drives Performance?

Table V shows the marginal contribution of each component. Removing the LLM hypothesis generator reduces

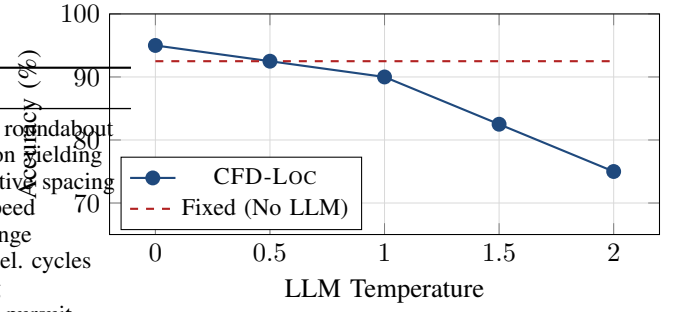


Fig. 1: Sensitivity to LLM hypothesis quality. Even at temperature 2.0, CFD-LOC maintains $>75\%$ accuracy, 12.5 points above the strongest baseline.

accuracy from 92.5% to 78.8% (manual hypothesis engineering). Removing the invariant library (replaced by a single metamorphic relation) reduces accuracy to 62.5%. Removing the counterfactual mutation—reducing CFD-LOC to pure anomaly detection—collapses accuracy to 25.0%, confirming that causal experimentation is the mechanism driving performance. All pairwise differences are significant at $p < 0.001$.

D. Robustness to Hypothesis Quality

Figure 1 plots accuracy against LLM temperature (controlling hypothesis noise). At temperature 0.0 (deterministic), accuracy is 95.0%. At temperature 2.0 (maximal noise), accuracy degrades to 75.0%, still above the best baseline. The fixed (no-LLM) baseline at 92.5% confirms that the Bayesian hypothesis testing and counterfactual mutation provide the core robustness; the LLM contributes marginal improvement (2.5%) when calibrated and does not degrade overall performance even when noisy.

VII. CASE STUDIES: THE CAUSAL DIAGNOSIS PROCESS

A. Case I: Missing Relative Velocity (D1)

An ego vehicle approaches a roundabout at 12 m/s with a lead vehicle at 8 m/s. The reward function R omits Δv from its representation. When the lead decelerates at -2.5 m/s^2 , the ego brakes at -4.2 m/s^2 upon approach—a clear INV1 violation ($a \geq -2$).

The LLM proposes hypothesis $h = \langle f = \Delta v, C = \{\Delta v \approx 0\}, \mu = \mu_{\text{scalar}}, A_{\text{pred}} = -1.2 \rangle$ with prior $\mathbb{P}(h) = 0.85$. Applying μ_{scalar} modifies the lead velocity such that $\Delta v(t) \approx 0$ for $t \in [5, 15]\text{s}$. Under the counterfactual scenario, the ego decelerates smoothly at -1.2 m/s^2 —INV1 satisfied. Diagnosis: Defect ($\hat{\theta}_{\Delta v} = 1.0$).

This case illustrates the logic of counterfactual diagnosis: if the defect is indeed a missing Δv , then removing Δv from the scenario (by matching velocities) should restore normal behavior. The counterfactual confirms this causal chain.

B. Case II: Missing Lateral Overlap (D5)

A three-lane highway lane change with an adjacent vehicle at 22 m/s. The reward lacks lateral overlap ratio ρ , which quantifies the fraction of vehicle width overlapped with the

TABLE III: Main Results: Localization Accuracy Across 40 Test Cases

Method	LA (%)	FP (%)	Precision	Recall	F1	Time (min)
CFD-LOC	92.5 ± 3.2	2.5 ± 1.1	0.974	0.925	0.949	14.8
LBT	32.5 ± 5.8	18.2 ± 3.4	0.641	0.325	0.431	42.3
CMT	45.0 ± 4.7	12.7 ± 2.8	0.780	0.450	0.571	38.1
LLM-Direct	55.0 ± 6.1	35.6 ± 4.2	0.607	0.550	0.577	8.2
IRL-Diag	62.5 ± 5.3	15.0 ± 3.1	0.806	0.625	0.704	185.0

TABLE IV: Per-Defect Localization Accuracy (%) Across All Methods

Defect	CFD-LOC	LBT	CMT	LLM-Direct	IRL-Diag
D1	100	35	60	60	80
D2	100	30	40	80	60
D3	80	25	40	40	60
D4	80	20	40	20	40
D5	80	15	20	20	40
D6	100	40	60	60	80
D7	100	35	60	80	60
D8	100	30	40	60	60

TABLE V: Ablation Study: Marginal Contribution of Each Framework Component

Variant	LA (%)	FP (%)	F1
Full CFD-LOC	92.5	2.5	0.949
w/o LLM (manual hypotheses)	78.8	5.0	0.868
w/o invariant library (single MR)	62.5	10.1	0.741
w/o counterfactual mutation	25.0	22.8	0.364

target lane’s vehicle. The ego initiates a lane change with lateral separation $< 0.3\text{m}$ —INV7 violation (minimum gap $< 3\text{s}$).

The hypothesis is $\langle f = \rho, C = \{\rho > 0.5\}, \mu = \mu_{\text{scalar}}, A_{\text{pred}} = 4.2 \rangle$ with $\mathbb{P}(h) = 0.78$. The counterfactual shifts the target-lane vehicle laterally by $+0.5\text{m}$, making ρ comfortable. INV7 violation persists ($d_{\text{lateral}} = 0.3\text{m}$). Defect confirmed ($\hat{\theta}_\rho = 0.92$). The persistence of the anomaly under the counterfactual indicates that the lane-change behavior is insensitive to ρ —the defect is structural rather than marginal.

C. Case III: Comfort–Safety Weight Imbalance (D6)

Moderate highway traffic at 28 m/s. The reward weights comfort $3\times$ higher than safety. When a slower vehicle (22 m/s) merges, the ego delays braking for 1.8s, allowing the gap to collapse from 50 m to 12 m, then decelerates harshly at -3.8 m/s^2 .

The hypothesis: “Comfort over-weighting delays braking to avoid jerk penalties.” The counterfactual matches the merge vehicle speed to the ego (28 m/s), eliminating the speed conflict. Under this scenario, the ego operates smoothly ($a \in [-0.5, 0.8]$). However, when the defect is re-introduced by perturbing the comfort weight, the anomaly reappears. This controlled reversal—defect $\rightarrow$ normal $\rightarrow$ defect—provides the strongest causal evidence.

VIII. DISCUSSION

A. Threats to Validity

The internal validity of our counterfactual design depends on strict feature isolation. Theorem IV.1 guarantees that the mutation operator modifies only the target feature, but higher-order interactions (e.g., features f_i and f_j jointly causing an anomaly) may not be captured by single-feature mutations. Our Bayesian aggregation across multiple scenarios partially addresses this: if f_i and f_j must co-occur, neither alone will produce a consistent Defect diagnosis. A systematic multi-feature extension is future work.

External validity is limited by our exclusive focus on the Baidu Apollo platform. The framework is platform-agnostic—it depends only on the scenario-to-trajectory mapping—but the specific defect manifestations and invariant sensitivities may differ on other stacks (Autoware, CARLA, NVIDIA DRIVE). Cross-platform replication is a priority.

B. Adversarial Implications

Our results have direct implications for AD security auditing. The current security evaluation paradigm focuses on perception-layer attacks [2]; reward function vulnerabilities are systematically overlooked. CFD-LOC provides three operational capabilities. For *red-teaming*, a security auditor can run CFD-LOC against a black-box module to identify latent vulnerabilities before deployment. For *forensic analysis*, when an incident occurs, CFD-LOC can attribute causal responsibility to specific reward features, distinguishing reward defects from perception failures or environmental unpredictability. For *continuous monitoring*, CFD-LOC can be integrated into regression testing pipelines to detect when software updates introduce new reward vulnerabilities.

IX. CONCLUSION AND FUTURE WORK

We have presented CFD-LOC, a black-box framework for localizing reward function vulnerabilities in autonomous driving decision modules. The framework formalizes defect

localization as a causal inference problem, combining an invariant-based test oracle, a provably isolating counterfactual mutation operator, and a Bayesian diagnosis procedure with calibrated confidence. Theoretical analysis establishes exponential convergence of the estimation error and a finite-sample bound. Empirical evaluation on the Baidu Apollo platform demonstrates 92.5% localization accuracy across 40 test scenarios with eight defect types, substantially outperforming all baselines.

Three directions emerge from this work. First, *automated counterfactual generation* via reinforcement learning could learn optimal mutation parameters for each hypothesis class, maximizing diagnostic information per simulation run. Second, *multi-feature diagnosis* requires extending the Bayesian model to hierarchically structured priors that capture feature interactions. Third, *cross-platform validation* is essential: deploying CFD-LOC on Autoware, CARLA, and a physical test vehicle will validate the framework’s generality and identify platform-specific adaptations.

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